EXPERIMENTAL CASE STUDY • DOCUMENTED 50% PROOF PREFIX

Frozen Prefix, Open Provenance

A Partially Blinded AMLD+0-Style Proof-Completion Attempt

The Collective-BANK Rendezvous Theorem (BC50-A)

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Research documentation, mathematical review, and typesetting assisted by ChatGPT

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Observed outcome: CANDIDATE COMPLETE. The model supplied a continuation that, on subsequent mathematical review, establishes the frozen theorem under the packet's explicit provenance, checker, proof-composition, and cost assumptions. No independent executable verifier accepted a concrete certificate, no authenticated six-role AMLD+0 runtime was executed, and the attempted blinding was *not* equivalent to a fresh, isolated session. This report is a reproducible record of a single mathematical attempt, not a verified formal-ATP benchmark.

"I have found AMLD+0 to be a more than adequate automated theorem prover."
— Brian Tenneson (author's experiential assessment; not an inference from this one trial)

Abstract

A prior frozen-prefix proof-completion study used the visible beginning of a theorem proof but could not rule out exposure to the original ending. This follow-up selected a different result from a 107-entry publication inventory: the Collective-BANK Rendezvous Theorem in *Ant-Colony Rendezvous for Verifier-Backed Bidirectional Proof Search*. The publicly archived 110-word source proof was mechanically split after 55 whitespace-delimited words. A two-page solver packet withheld the article identity and proof ending, supplied the theorem and the requisite assumptions, and asked for a continuation through an "unarmed" AMLD+0-style prompt. The observed ChatGPT response gave a mathematically coherent continuation and assembled proof, marked CANDIDATE COMPLETE, and correctly declined to claim independent formal verification. Post-run comparison found agreement on the theorem's central proof composition and distance bound; the original hidden suffix also stated an optional disjoint-union upper bound that the solver omitted. Because the prompt was ultimately submitted in the continuing preparation conversation, with earlier source access in that context, the experiment must be classified as *partially blinded*. Full original packet and organizer instructions are preserved within this single PDF.

1. Why this experiment was conducted

The research question is narrow: can a designated solver continue a *specific finite proof* from a frozen visible prefix, rather than merely assert that some proof exists? An earlier example concerned the Induction Principle on Proofs in *Depths of a Simulation 45.2*. Its mathematical continuation was informative, but source exposure in the same conversation prevented a strong blind-discovery claim [2]. The present test sought a clearer information boundary: a newly selected theorem, a precise 50% word-count cut, a source-free packet, and a declared audit rubric.

The name AMLD+0 denotes the author's intended conjecture-settling architecture, with global Player-0 and settlement roles P, R, D, I, C. In this case, the solver received an AMLD+0-*styled instruction*. That alone

does not demonstrate a six-agent scheduler, frozen verifier executable, or trusted BANK implementation. This distinction is central to the finding.

2. Source selection and frozen record

The user supplied *Brian Tenneson Library Inventory 10-22-2026.xlsx*, containing **107 numbered publication entries**. The candidate was deliberately selected rather than randomized:

> **Inventory entry 77:** *Ant-Colony Rendezvous for Verifier-Backed Bidirectional Proof Search*, revised working theorem draft 1.2 (inventory difficulty rating 8/10, article level).
> **Chosen result:** *Collective-BANK Rendezvous Theorem*.
> **Why chosen:** it tests multi-source certified proof composition and is not another Exact Addressing 4.14 or Depths 45.2 proof. It has a substantive dependency-merging argument despite its short written proof.

The source was frozen against a Git repository record before the attempt:

- Repository: `btenneson/pub`; branch snapshot commit `7b398f052970735d75fc8a39065701bb6d1d9899`.
- Relative source: `docs/cs.LO_Logic_in_Computer_Science/ant_colony_rendezvous_verifier_bidirectional_proof_search_v1_2/source.tex`.
- Git blob SHA: `5923edd2c9a11ec69d9354da49414e2f63b8ff5c`.
- The selected proof contains 110 whitespace-delimited LaTeX-source words; the first 55 were exposed and the final 55 withheld in the solver packet. *Word fraction is not a formal proof-step fraction.*

The article was identifiable to the organizer; its title, author, URL, source-path manifest, and original conclusion were not included in the *solver-only* packet. Thus "blind" describes the packet's intended information barrier, not a guarantee about model training or all chat context.

3. The frozen theorem and permitted mathematical interface

The packet specifies a presented verifier-backed system $F = (\Sigma, \Gamma, R, V, w)$ with finitary rules, nonnegative primitive costs, and initial trusted state $B_0 = \Gamma \cup \mathrm{Ax}(F)$. A backward probe $\mathfrak{b} = (O, \sigma^-)$ comprises a finite obligation set and certified suffix, so that

$$B_0 \cup O \vdash_F T, \qquad g^-(\mathfrak{b}) = \mathrm{cost}(\sigma^-).$$

A monotone global BANK B_t is *provenance-complete* if every entry has a finite verifier-replayable justification from B_0. The object $D(O)$ is the authorized, finite, acyclic merge of the provenance DAGs for O, accounting for valid DAG sharing. The distance $d_F(B_0, T)$ is the infimum charged cost of legal verified paths to a state containing T. Proof replay and composition are assumed to preserve rules, scope, verifier acceptance, and the additive upper-bound convention.

Collective-BANK Rendezvous Theorem. If $\mathfrak{b} = (O, \sigma^-)$ is a backward obligation probe and $O \subseteq B_t$ for a provenance-complete monotone global BANK, then $D(O)$ followed by σ^- yields a finite verifier-replayable proof of T. Moreover

$$\boxed{d_F(B_0, T) \leq \mathrm{cost}(D(O)) + g^-(\mathfrak{b}).}$$

The obligations in O may have been established by different agents or along different forward search trajectories.

3.1 Exact visible proof prefix

The public source was divided after the 55th whitespace-delimited word. The actual packet displayed the following LaTeX prefix, ending mid-sentence:

> For every $P \in O$, provenance completeness supplies either the trivial zero-step certificate when $P \in B_0$ or a finite verifier-accepted derivation DAG D_P from B_0 to P. Since O is finite, an allowed provenance merge $D(O)$ is finite, acyclic, and topologically replayable from B_0, establishing every member of O. The stored suffix σ^- is then replayable

The exact original solver-packet PDF (two pages, including all supporting definitions) is embedded unchanged in Appendix C.

4. Protocol, chronology, and information flow

1. Organizer selection: inventory entry 77; locate selected theorem; freeze public source commit/blob. Organizer has source access.

↓

2. Packet construction: copy theorem and assumptions; release 55/110 words; withhold 55/110; omit article identifiers.

↓

3. Actual submission: user attaches packet and pastes the longer AMLD+0 prompt *in the same ongoing conversation*, not a fresh Temporary Chat.

↓

4. Observed answer: ChatGPT supplies candidate suffix, assembled proof, caveats, and label CANDIDATE COMPLETE. No external verifier receipts.

↓

5. Post-attempt evaluation: compare mathematical content with frozen original ending; document ambiguity of isolation and lack of authentic AMLD+0 run.

4.1 What was planned versus what actually occurred

The organizer PDF recommended a fresh Temporary Chat, supplying *only* the solver packet, with external retrieval disabled. The observed submission instead occurred in the continuing conversation that designed the task and had previously fetched and inspected the public source. The user pasted a longer "unarmed" prompt, reproduced in Appendix A; it was *not* the shorter alternative prompt distributed as a separate text file. The preparation-stage source access is a live contamination concern even though the candidate answer itself did not quote the hidden ending.

No predeclared CPU/service-turn budget, randomized seed, detailed wall-clock record, or verified isolation receipt was preserved. No executed simulation with a frozen run log, and no separate P, R, D, I, C agent transcripts, were supplied. The actual object under evaluation is one ChatGPT candidate mathematical response framed by AMLD+0 rules.

5. Observed proof-completion response

The solver's emitted label was **CANDIDATE COMPLETE**. Its missing-suffix answer began exactly after "The stored suffix σ^- is then replayable":

> from the now-established obligations O. Indeed, every obligation required by the suffix is either already in B_0 or has a verifier-replayable derivation in $D(O)$. Substituting these derivations for the corresponding open premises of σ^- produces a finite derivation from B_0 to T. By the assumed sound proof-composition interface, the complete derivation is verifier-replayable and retains the original assumption scope. Its charged cost is at most the cost of the merged provenance followed by the suffix:
>
> $$\mathrm{cost}(\pi_T) \le \mathrm{cost}(D(O)) + g^-(\mathfrak{b}).$$
>
> Since $d_F(B_0, T)$ is the infimum over legal verified paths reaching T, it cannot exceed the cost of this constructed path. The individual provenance DAGs may originate from different agents or histories; their common certified base B_0 permits their legal composition. This proves the result.

5.1 Assembled mathematical argument

The solver then presented the following complete proof, using the original prefix and its continuation (typeset from the actual answer without mathematical repair):

> For every $P \in O$, provenance completeness supplies either the trivial zero-step certificate when $P \in B_0$ or a finite verifier-accepted derivation DAG D_P from B_0 to P. Since O is finite, an allowed provenance merge $D(O)$ is finite, acyclic, and topologically replayable from B_0, establishing every member of O.
>
> The stored suffix σ^- is then replayable from the now-established obligations O. Its open premises are contained in $B_0 \cup O$. Each premise in O can therefore be discharged by its certified derivation contained in $D(O)$.
>
> Substitute those derivations into the suffix. Because O is finite and the suffix contains finitely many inference steps, this substitution produces a finite proof DAG whose leaves are justified by B_0 and whose conclusion is T.
>
> The frozen proof-composition interface preserves the calculus, assumption scopes, and verifier rules. Consequently, the composed proof π_T is verifier-replayable and establishes $B_0 \vdash_F T$.
>
> By the cost-composition assumption,
>
> $$\mathrm{cost}(\pi_T) \le \mathrm{cost}(D(O)) + \mathrm{cost}(\sigma^-).$$
>
> Using $g^-(\mathfrak{b}) = \mathrm{cost}(\sigma^-)$, we have
>
> $$\mathrm{cost}(\pi_T) \le \mathrm{cost}(D(O)) + g^-(\mathfrak{b}).$$
>
> The constructed proof provides a legal verified path from B_0 to a state containing T. By the definition of intrinsic target distance as an infimum,
>
> $$d_F(B_0, T) \le \mathrm{cost}(\pi_T) \le \mathrm{cost}(D(O)) + g^-(\mathfrak{b}).$$
>
> Finally, nothing requires the certificates in $D(O)$ to originate from one agent or one forward search history. Their common, verifier-accepted provenance from B_0, together with the authorized merge interface, is sufficient. □

5.2 The solver's explicit qualifications

The solver gave two supporting lemmas in prose: finite provenance composition and certified substitution of the derived obligations into the suffix. It did not introduce new premises. The answer expressly stated that no concrete DAGs D_P or executable suffix certificate were available for replay, no implementation of V was provided, and no authenticated Player-0/role/service-cost transcript existed. Its own verification table (reproduced in Appendix B) said "Independent executable verifier: Not run" and "Actual AMLD+0 federation: Not run." It classified the result as CANDIDATE COMPLETE, not certified PROVED.

6. Post-attempt comparison with the frozen reference

Only after the answer had been emitted did the present evaluation retrieve the frozen original continuation for direct comparison. The 55 withheld whitespace-delimited source words were:

> because its open obligations are available. Appending the suffix to the replay of $D(O)$ therefore gives a verifier-accepted proof of T. The distance bound follows because a concrete proof of the displayed realized cost has been exhibited. In particular, using the disjoint provenance union gives the always-safe coarser bound
>
> $$d_F(B_0, T) \le \sum_{P \in O} \mathrm{cost}(D_P) + g^-(\mathfrak{b}).$$

Both texts invoke the same central mechanism: proof DAGs for each obligation are combined into a provenance-complete merge; the stored suffix is replayed after those obligations are established; valid composition produces T; and this witnessed proof upper-bounds $d_F(B_0, T)$. The candidate explicitly discusses scope-preserving substitution and the independence of agent histories, rather than merely echoing the source prose.

Criterion	Reference suffix	Candidate continuation
Open obligations discharged	Explicit	Explicit, via substituted subproofs
Verifier-replayable composite	Explicit	Explicit, conditional on frozen interface
Primary target-distance bound	Explicit	Explicit
Different producing agents allowed	Implied by theorem; source setting	Explicit
Optional disjoint-union bound	Explicit	Not stated
Externally checked proof object	Not part of short source derivation	Not provided

The omitted *coarser* disjoint-union estimate is supplementary to the theorem as stated; it is not a logical gap in the main completion. Conversely, matching the source's proof strategy does not establish independent discovery, because prior model exposure and same-session contamination remain uncontrolled.

7. Assessment and limits

Mathematics: The candidate contains a complete abstract proof under the stated finite-provenance, scope, proof-composition, and cost conditions. A post-hoc mathematical review found no unresolved inference *within that abstract setting*.
Provenance and execution: The response is not a fully blind solver trial, not an authenticated AMLD+0 federation run, and not a formally checked theorem certificate.
Appropriate reported outcome: CANDIDATE COMPLETE (POST-HOC HAND-REVIEWED); INDEPENDENT FORMAL VERIFICATION NOT PERFORMED; INFORMATION BARRIER INCOMPLETE.

There are at least six concrete threats to stronger conclusions:

1. **Prior exposure.** The source article was publicly available and accessed during preparation in the same conversation; model-training exposure cannot be excluded.
2. **Session leakage.** The fresh Temporary Chat prescribed by the organizer was not used. The packet may have omitted identifiers, but the surrounding context was not provably isolated.
3. **Short theorem.** A 110-word original proof can express a routine compositional consequence of strong explicit definitions; the 55-word prefix already supplies the key provenance construction.
4. **Mathematical abstraction.** The case establishes a conditional derivation, not the successful output of an executable ATP on concrete inference objects.
5. **No independent checker.** There is no machine-verifier acceptance receipt and no independent reviewer transcript. The present comparison was conducted by the same assistant product.
6. **Single selected case.** One nonrandom trial cannot estimate success rate, speedup, robustness, novelty, or global capability of AMLD+0.

The first-person sentence "I will finish the proof" was operationally satisfied only in the limited sense that the chat produced a candidate continuation. Without the declared runnable verifier, bound, and authenticated federation logs, it is *not* established as the stronger historical AMLD+0 first-person verified-success event.

8. Improved protocol for a genuinely stronger follow-up

To turn this case into a prospective test of a particular AMLD+0 implementation, an independent organizer should: (i) select a never-published or newly authored theorem and seal a source hash; (ii) freeze the exact proof-language interface and permissible BANK; (iii) establish a clean isolated solver environment before any theorem-source access, including connected-app and web restrictions; (iv) choose the prefix boundary by declared proof steps as well as words; (v) freeze engine hashes, six-role identities, Player-0 policy, random seed, service/verification cost ledger, and a stopping budget; (vi) archive the entire execution transcript and certificate; and (vii) obtain checker acceptance before revealing the reference continuation. Success and failure should be reported under the same rule.

This test still contributes a concrete methodological specimen: the prepared packet, real user prompt, full candidate argument, source-comparison basis, and documented departure from intended isolation are all archived below, in a single publication file.

9. Conclusion

At the formal-mathematical level, the generated suffix completed the Collective-BANK Rendezvous proof. It recognized that multiple provenance DAGs can be combined to discharge the obligations of a certified backward suffix, thereby deriving the target and its associated cost upper bound. The improvement over the earlier frozen-prefix trial lies in explicit source identification, a 55/110 prefix boundary, an unchanged solver packet, and stronger disclosure of assumptions and evidence. The remaining limitation is fundamental: *the actual trial did not establish complete blindness or a real AMLD+0 execution*. A careful publication can report both the constructive mathematical result and that methodological limitation without confusing either one for the other.

Sources and provenance

References

[1] Brian Tenneson, *Ant-Colony Rendezvous for Verifier-Backed Bidirectional Proof Search*, revised working theorem draft 1.2 (September 2026), https://btenneson.github.io/pub/cs.LO_Logic_in_Computer_Science/ant_colony_rendezvous_verifier_bidirectional_proof_search_v1_2/. Git snapshot, file hash, and theorem identified in §2.

[2] Brian Tenneson, *AMLD+0 Frozen-Prefix Proof Completion: Depths 45.2*, earlier experimental article, https://btenneson.github.io/pub/cs.LO_Logic_in_Computer_Science/amld0_frozen_prefix_proof_completion_depths_45_2_0_1/.

[3] Brian Tenneson, *Library Inventory 10-22-2026*, user-provided spreadsheet; numbered entry 77.

[4] *Blind_50pct_Solver_Packet_BC50A.pdf*, frozen two-page challenge supplied to the solver, reproduced on original pages in Appendix C.

[5] *Blind_50pct_Organizer_and_Article_Plan_BC50A.pdf*, organizer-only three-page method guide, reproduced in Appendix D.

[6] User's actual longer AMLD+0 prompt and the assistant's subsequent candidate proof, recorded in the conversation on 23 October 2026; transcribed in this report.

Appendix A. Actual submitted prompt (verbatim)

The text below is the *longer instruction actually pasted by the user*. It differs from the shorter plain-text solver prompt prepared earlier. Its original Markdown formatting and sequence are retained, with line wrapping for print.

```
AMLD+0, please settle the following conjecture:

**I will finish the proof.**

**M:** Your designated AMLD+0 system, with Player-0 and settlement roles P, R, D, I, C.

**F, H, T, V:** Exactly as specified in the attached blind proof-completion packet.

**BANK K0:** Only the admitted definitions, hypotheses, theorem statement, and visible 50% proof
↪  prefix supplied in that packet.

**Target:** Construct a complete, mathematically valid continuation of the unfinished proof without
↪  changing its theorem or assumptions.

**Frozen conditions:**

1. You are unarmed. No source book, original article, reference proof, external ATP, web search,
↪  connected applications, or additional proof libraries are supplied or authorized.
2. Do not search for, request, or reconstruct the hidden reference text by looking it up.
3. Continue from the exact point where the visible proof ends.
4. Player-0 may coordinate the declared settlement roles but may not change the theorem, hypotheses,
↪  verifier, or acceptance conditions.
5. You may introduce intermediate lemmas only when you justify them from the admitted premises and
↪  rules.
6. Preserve all proof dependencies and assumption scopes.
7. Do not substitute a simulator or another proof engine without authorization.
8. If no independently executable verifier or multi-agent runtime is available, clearly identify the
↪  result as a mathematical candidate rather than an authenticated AMLD+0 execution.
9. Do not assume the theorem is proved merely because the original proof has a continuation.
10. UNKNOWN is permitted. Do not invent a proof.

**Required output:**

A. Your proposed missing 50% of the proof.
B. The complete assembled proof, including the supplied prefix.
C. Any auxiliary lemmas and their justifications.
D. An explicit list of unverified steps or remaining obligations.
E. Whether an independent verifier actually accepted the result, with receipts if available.
F. Final result: CANDIDATE COMPLETE, INCOMPLETE, or UNKNOWN.

**First-person requirement:**

The statement “I will finish the proof” refers to your own actual attempt. A possible proof path is
↪  not sufficient. You must produce the continuation and submit the resulting proof.

Proceed using only the attached packet.
```

Appendix B. Candidate answer: full A–F content transcript

The following preserves the substantive content and sequence of the actual solver response, including its own evidence qualifications and result table. Mathematical symbols have been normalized to printable plain text; the typeset mathematical proof in §5 above retains its exact logical claims. This is a documentary transcription, not an independent verifier receipt.

```
CANDIDATE COMPLETE

I have completed the missing argument using only the supplied blind solver packet BC50-A. The
↪  continuation establishes the theorem under its frozen assumptions.

This is a hand-checked mathematical proof candidate-not an authenticated execution of the six AMLD+0
↪  roles, and not an independently machine-verified certificate.

A. Proposed missing 50%

The supplied prefix ends:

> The stored suffix \sigma^- is then replayable

My continuation begins at that exact point:

> from the now-established obligations O. Indeed, every obligation required by the suffix is either
↪  already in B_0 or has a verifier-replayable derivation in D(O). Substituting these derivations
↪  for the corresponding open premises of \sigma^- produces a finite derivation from B_0 to T. By
↪  the assumed sound proof-composition interface, the complete derivation is verifier-replayable and
↪  retains the original assumption scope. Its charged cost is at most the cost of the merged
↪  provenance followed by the suffix:
>
> cost(\pi_T) <= cost(D(O)) + g^-(b).
>
> Since d_F(B_0,T) is the infimum over legal verified paths reaching T, it cannot exceed the cost of
↪  this constructed path. The individual provenance DAGs may originate from different agents or
↪  histories; their common certified base B_0 permits their legal composition. This proves the
↪  result.

That is my proposed continuation, not a quotation from the hidden reference.

B. Complete assembled proof

Theorem. Let b=(O,\sigma^-) be a backward obligation probe satisfying B_0 union O |-F T. Suppose O is
↪  a subset of B_t, where B_t is a provenance-complete monotone global BANK. Then D(O), followed by
↪  \sigma^-, provides a finite verifier-replayable proof of T. Consequently,

    d_F(B_0,T) <= cost(D(O)) + g^-(b).

Proof. For every P in O, provenance completeness supplies either the trivial zero-step certificate
↪  when P in B_0 or a finite verifier-accepted derivation DAG D_P from B_0 to P.

Since O is finite, an allowed provenance merge D(O) is finite, acyclic, and topologically replayable
↪  from B_0, establishing every member of O.

The stored suffix \sigma^- is then replayable from the now-established obligations O. Its open
↪  premises are contained in B_0 union O. Each premise in O can therefore be discharged by its
↪  certified derivation contained in D(O).

Substitute those derivations into the suffix. Because O is finite and the suffix contains finitely
↪  many inference steps, this substitution produces a finite proof DAG whose leaves are justified by
↪  B_0 and whose conclusion is T.
```

```
The frozen proof-composition interface preserves the calculus, assumption scopes, and verifier rules.
↪  Consequently, the composed proof \pi_T is verifier-replayable and establishes B_0 |-F T.

By the cost-composition assumption,

    cost(\pi_T) <= cost(D(O)) + cost(\sigma^-).

Using the definition g^-(b) = cost(\sigma^-), we have

    cost(\pi_T) <= cost(D(O)) + g^-(b).

The constructed proof provides a legal verified path from B_0 to a state containing T. By the
↪  definition of the intrinsic target distance as an infimum,

    d_F(B_0,T) <= cost(\pi_T) <= cost(D(O)) + g^-(b).

Finally, nothing requires the certificates in D(O) to originate from one agent or one forward search
↪  history. Their common, verifier-accepted provenance from B_0, together with the authorized merge
↪  interface, is sufficient.

QED.

C. Auxiliary lemmas

No new mathematical assumptions were introduced. The argument relies on two elementary consequences
↪  of the packet's stated assumptions.

Finite provenance composition. A finite family of certified finite DAGs rooted in the same trusted
↪  base admits a verifier-replayable merge when the declared merging interface permits it.

Certified substitution. If B_0 |-F P for every P in O, and a certified suffix establishes B_0 union O
↪  |-F T, sound proof composition establishes B_0 |-F T.

Both are applications of the proof-composition assumptions supplied in BC50-A.

D. Remaining verification obligations

The abstract mathematical argument appears complete, but these concrete matters have not been
↪  independently tested:

- No actual proof DAGs D_P or executable suffix certificate were provided for replay.
- No implementation of the frozen verifier V was supplied.
- The cost inequality is conditional on the packet's declared cost-composition interface.
- No authenticated Player-0 schedule, role transcript, or AMLD+0 execution log exists for this
↪  response.

There is also an important experimental caveat: this is not a fully isolated blind environment,
↪  because earlier parts of this conversation involved preparing the challenge. I did not consult
↪  the hidden continuation while producing this answer, but that fact alone cannot establish
↪  model-level ignorance.

E. Verification record

Check                                Result
Original theorem preserved           Yes
Visible prefix preserved             Yes
Missing argument supplied            Yes
Assumption scopes addressed          Yes, conditionally
Cost bound established               Yes, conditionally
Independent executable verifier      Not run
```

```
Actual AMLD+0 federation              Not run
Hidden reference comparison           Not performed

F. Final settlement result

BC50-A - 50% proof-completion challenge
CANDIDATE COMPLETE

The missing continuation has been constructed, and the abstract mathematical argument is complete
↪  under the supplied hypotheses. External certificate verification remains outstanding.

For the experiment record: I would classify this as a successful candidate proof completion, with
↪  independent formal acceptance unconfirmed-not as a certified AMLD+0 PROVED result.

The next step is to preserve this answer unchanged, then give the complete response to the experiment
↪  organizer for comparison with the sealed reference proof. That comparison should assess
↪  mathematical correctness independently of whether the wording matches the original.

No hidden reference proof was consulted or revealed in this response.
```

Appendix C. Original solver packet (exact two-page facsimile)

The following pages reproduce the actual submitted solver-only packet unaltered, including the theorem, definitions, and 55-word prefix. Any article identifiers in this publication report were added *after* the attempt for readers; they were absent from these pages.

BLIND PROOF COMPLETION

Solver-only packet | Challenge BC50-A | 50% visible prefix | No source identifiers

Task and trust boundary

The task is to complete the missing portion of a finite mathematical proof, continuing from the frozen visible prefix. The original reference proof, publication title, URL, author, and proof ending are deliberately omitted. Do not consult external sources or other conversations. This is a source-isolated mathematical challenge; it is not proof of permanent ignorance or an authenticated multi-agent execution.

Frozen formal setting

Let $F = (\Sigma, \Gamma, R, V, w)$ be a presented formal system with finitary inference rules, a verifier V for finite certificates, and nonnegative primitive proof-step costs w. The initial trusted state is $B_0 = \Gamma \cup Ax(F)$. An accepted certificate at another scope may not be imported unless the assumptions are justified and the required proof composition is checked.

Permitted definitions

Backward probe. A pair $b = (O, \sigma^-)$, where O is a finite set of open proof obligations and σ^- is a finite certified suffix schema establishing $B_0 \cup O \vdash_F T$. Its recorded suffix cost is $g^-(b) = cost(\sigma^-)$.

Provenance-complete BANK. A monotone trusted BANK B_t is provenance-complete if every P in B_t has a finite verifier-replayable provenance certificate from B_0. An initial premise may use the trivial zero-step certificate. For finite $O \subseteq B_t$, define $D(O)$ to be a finite acyclic, verifier-replayable merge of the provenance DAGs for obligations in O. Its cost is the declared realized replay cost, counting shared nodes only as permitted by the verifier and accounting convention.

Target distance. Write $d_F(B_0,T)$ for the infimum cost of a legal verified path from B_0 to a proof state containing T. Assume that the replay/composition interface preserves the frozen calculus, assumptions and verifier, and that the composed cost is bounded by the sum of the stated component costs.

Theorem to prove

Let $b = (O, \sigma^-)$ be a backward obligation probe with $B_0 \cup O \vdash_F T$. If $O \subseteq B_t$ for a provenance-complete monotone global BANK B_t, then the dependency merge $D(O)$, followed by the certified suffix σ^-, yields a finite verifier-replayable proof of T. Consequently:

$$d_F(B_0,T) \leq cost(D(O)) + g^-(b).$$

The obligations in O need not have been established along one forward trajectory or by one forward agent.

Visible prefix of the original proof

Exactly 55 of 110 whitespace-delimited source words are shown. The cut is mechanical and ends mid-sentence. No original continuation is included.

```
For every $P\in O$, provenance completeness supplies either the trivial
zero-step certificate when $P\in B_0$ or a finite verifier-accepted
derivation DAG $D_P$ from $B_0$ to $P$. Since $O$ is finite, an allowed
provenance merge $D(O)$ is finite, acyclic, and topologically replayable
from $B_0$, establishing every member of $O$. The stored suffix $\sigma^-$
 is then replayable
```

Required solver output

Continue at the cut. Submit a mathematically sound continuation, then an assembled proof with any necessary intermediate lemmas or scope checks. Explicitly label which claims are hand-reasoned and whether an independent checker was used. If unable to complete the proof, report INCOMPLETE or UNKNOWN; do not manufacture acceptance receipts or call noncompletion a refutation.

Appendix D. Frozen organizer plan (exact three-page facsimile)

The following pages are included for methodological transparency. They are *organizer-only*: the recommended separate Temporary Chat was not used during the actual candidate answer. Do not interpret these planned actions as a record of actions performed.

ORGANIZER & PUBLICATION PLAN

Separate from solver | Challenge BC50-A | Reference proof sealed in frozen public-source snapshot

What has been selected

Candidate pool: October 22, 2026 library inventory, publication #77. Source article: Ant-Colony Rendezvous for Verifier-Backed Bidirectional Proof Search, revised working draft 1.2. Selected result: Collective-BANK Rendezvous Theorem. The original proof has 110 whitespace-delimited source words; the solver-only packet contains the first 55. The hidden 55 words have not been placed in the solver packet.

Frozen source manifest

Repository: btenneson/pub; branch snapshot commit: 7b398f052970735d75fc8a39065701bb6d1d9899. Source file at that commit:

```
docs/cs.LO_Logic_in_Computer_Science/
ant_colony_rendezvous_verifier_bidirectional_proof_search_v1_2/
source.tex
```

Git blob SHA of source.tex: 5923edd2c9a11ec69d9354da49414e2f63b8ff5c. The organizer/evaluator can retrieve the original continuation later using this frozen manifest. Do not give this page, file path, article title, or the source to the solver.

Step-by-step: what to do, and when

1. Now - stop preparation

Keep the solver-only challenge unchanged. Do not inspect or paste the hidden continuation into the solver conversation.

2. When rested - open a fresh isolated chat

Prefer a Temporary Chat or an equivalent session without remembered prior conversation. Do not name the article or its URL. Do not add public library links or the organizer PDF.

3. Give the solver one packet and one prompt

Attach Blind_50pct_Solver_Packet_BC50A.pdf. Paste the accompanying plain-text prompt Blind_50pct_Solver_Prompt_BC50A.txt unchanged. Turn web search, connected sources, and external proof retrieval off where possible.

4. Freeze and record the run

Record date/time, exact model/configuration, starting prompt and packet version, declared budget, any tools actually used, transcript, output, and verification receipts. A plain ChatGPT answer is not a verified six-role AMLD+0 run.

5. Return here for evaluation

Bring the entire solver response as text, PDF, or transcript plus any independently checkable proof object and verifier log. Send only the completed output and actual tool records, not a success claim.

6. Grade independently, then compare with the reference

First judge correctness of the submitted proof against the declared calculus and scope; ideally pass through an independent formal checker. Next retrieve the sealed reference ending and compare mathematical content, missing obligations, originality limitations, and provenance.

7. Write up success OR failure

Either outcome can be a publishable methodological case study if accurately reported. Report pass/fail/indeterminate, error types, amount of source exposed, contamination risks, and all changes from the prior 45.2 study.

GRADING & ARTICLE OUTLINE

Predeclared criteria and cautious reporting

Outcome classification

PASS - hand-checked: a complete valid continuation in the frozen calculus and scope, with no unjustified inference, but no independent checker receipt. PASS - externally checked: the same, plus independent acceptance of a full formal proof. INCOMPLETE: meaningful partial progress without a complete proof. FAIL: invalid derivation or scope breach. INDETERMINATE: verifier, source, or transcript ambiguity prevents reliable grading. A match with the original prose alone is not sufficient for PASS.

Proof comparison and information barrier

Do not judge solely by verbatim agreement: a valid alternative completion may differ from the source. Track whether the solver searched the public article, used connected files, received the title, relied on prior conversation, or claimed external verification that was not performed. This session and the model family cannot be certified never to have encountered the reference text; the goal is the enforced source barrier during the solver run.

Publishable report structure

- Abstract: successful, failed, or inconclusive result stated prominently.
- 1. Motivation: contrast the earlier 45.2 exposed-ending experiment with this isolated challenge.
- 2. Frozen source and fair selection: article pool, selection method, snapshot commit and proof length.
- 3. Challenge design: exact theorem, supporting definitions, 55/110 cut, solver prompt and forbidden sources.
- 4. Execution record: complete transcript, model/configuration, allowed tools, timing and charge ledger.
- 5. Submission: solver continuation and assembled proof, reproduced without editorial repair.
- 6. Independent judgment: checker record or annotated hand-check, and identified gaps.
- 7. Reference comparison: restore original hidden ending only after judging submission.
- 8. Threats to validity: prior model exposure, choice of theorem, short reference proof, source isolation.
- 9. Conclusion: what the result does and does not demonstrate about AMLD+0 and ATP proof completion.

Scientific caution

A new chat prompted in AMLD+0 style is a proof-completion attempt by that model, not automatically an execution of the historical six-role AMLD+0 architecture. To claim a true AMLD+0 run, preserve authenticated Player-0 scheduling, five role traces, fixed BANK/verifier/budgets and checker receipts. An observed success is about this particular theorem and frozen context, not an unconditional general speedup.